

Static Replication for Dual-Directional Buffered PLUS on SP 500

1 Summary

The objective of this project is to price the Dual-Directional Buffered PLUS instrument by the static option replication approach. We use S&P 500 Index as the underlying asset and feature a digital-type payoff.

From the file, this product has pricing date on January 17, 2025 (T_0), issue date on January 23, 2025 (T_1), valuation date on January 29, 2027 (T_2) and maturity date February 3, 2027 (T_3) with initial index value 5996.66 which is the index closing value on the pricing date T_0 . Using OVME command in Bloomberg, we set underlying as SPX Index, trade on January 17 (T_0), 2025 at 4 p.m., settle on January 17, 2025 (T_0), vol as historical. It gives an estimated volatility of 12.543% and dividend rate of 1.327%. From OIS term structure, we extract discount rate from January 17, 2025 (T_0) to January 22, 2027 is 0.920675 and to January 24, 2028 is 0.883969, discount rate from January 23, 2025 (T_1) to January 27, 2027 is 0.921108 and to January 27, 2028 is 0.884590. We use linear interpolation to calculate discount rate between January 17, 2025 (T_0) and January 29, 2027 (T_3) is 0.91997489, discount rate between January 23, 2025 (T_1) and February 23, 2027 (T_3) is 0.91840667. It is reasonable to use a continuous dividend yield because it can be assumed that the dividends of the many constituent stocks of the index are uniformly distributed.

The estimated value of the structured product of Buffered PLUS, calculated using these data, is:

$$\hat{V} = 986.3148567486197 \quad (1)$$

2 Model Formulation

2.1 Instrument Payoff Structure

$$V(S_T) = \begin{cases} 1175, & \text{if } S_T \geq 1.16667S_0 \\ 1000 + 1000 \times 1.5 \times \frac{(S_T - S_0)}{S_0}, & \text{if } S_0 < S_T < 1.16667S_0 \\ 1000 + 1000 \times \left| \frac{S_T - S_0}{S_0} \right|, & \text{if } 0.9S_0 \leq S_T \leq S_0 \\ 1000 \times \frac{S_T}{S_0} + 100, & \text{if } S_T < 0.9S_0 \end{cases} \quad (2)$$

where:

- S_T is the underlying asset price at maturity.
- S_0 is the Adj Close price on January 17, 2025.

The buffer absorbs moderate losses down to 90% of S_0 . The upside participation rate is 1.5x, subject to a cap.

2.2 Risk-Neutral Valuation

Using a risk-neutral pricing framework, we discount the expected payoff under the risk-neutral probability measure $\mathbb{Q}$:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[V(S_T)] \quad (3)$$

where: r is the risk-free rate. $\mathbb{E}^{\mathbb{Q}}[\cdot]$ denotes the expectation under the risk-neutral measure.

Specifically, in discount part, our calculation of e^{-rT} is from issue date January 23, 2025 (T_1) to maturity date February 3, 2027 (T_3), in valuation part, our calculation of $V(S_T)$ is from pricing date January 29, 2027 (T_2) to maturity date February 3, 2027 (T_3).

2.3 Static Replication Approach

To accurately value the Buffered PLUS instrument, we collected key financial data from **Bloomberg**, ensuring that our model reflects current market conditions:

- **Risk-Free Rate (r):** Derived from the US Treasury yield curve and interpolated from the OIS term structure as of **[January 17, 2025]**.
- **Historical Volatility:** Extracted from Bloomberg's option market data from OVME for the S&P 500 Index on **[January 17, 2025]**.
- **Dividend Yield (q):** Sourced from Bloomberg's dividend yield estimates from OVME for the S&P 500 Index as of **[January 17, 2025]**.
- **Discount Factors:** Interpolated from Bloomberg's OIS term structure on **[January 17, 2025]** to compute risk-free discount rates.

2.3.1 Replication Strategy

To statically replicate the Buffered PLUS, we use a portfolio consisting of:

- Buy a zero-coupon bond to provide principal protection.
- Long European call option (ATM) with a 1.5x leverage factor to capture the enhanced upside.

- Short European call option at the cap level to limit the maximum payout.
- Long European put option (ATM) to create the buffer zone where losses are mitigated.
- Short put digital option (below the buffer threshold) to capture the jump in the payoff.
- Short put option (below the buffer threshold) to allow losses beyond the buffer.

2.3.2 Mathematical Formulation

The replication portfolio $V_{\text{rep}}(S_T)$ consists of weighted options and a bond:

- Buy a zero-coupon bond with face value:

$$FV_B = 1000 \quad (4)$$

- Long call option:

$$n_1 = \frac{1000 \times 1.5}{S_0}, \quad K_1 = S_0 \quad (5)$$

- Short call option:

$$n_2 = \frac{1000 \times 1.5}{S_0}, \quad K_2 = 1.16667S_0 \quad (6)$$

- Long put option:

$$n_3 = \frac{1000}{S_0}, \quad K_1 = S_0 \quad (7)$$

- Short put digital option:

$$n_4 = 100, \quad K_3 = 0.9S_0 \quad (8)$$

- Short put option:

$$n_5 = \frac{2 \times 1000}{S_0}, \quad K_3 = 0.9S_0 \quad (9)$$

2.4 Black-Scholes Option Pricing

The european option payoff is given by:

$$C_0 = e^{-r_{1,3}(T_3-T_1)} [S_0 e^{(r_{0,2}-d)(T_2-T_0)} N(d_1) - K N(d_2)]$$

$$P_0 = e^{-r_{1,3}(T_3-T_1)} [K N(-d_2) - S_0 e^{(r_{0,2}-d)(T_2-T_0)} N(-d_1)]$$

where $r_{1,3}$ is interest rate from T_1 to T_3 and $r_{0,2}$ is interest rate from T_0 to T_2 ,

$$d_1 = \frac{\log \frac{S_0}{K} + (r_{0,2} - d + \frac{1}{2}\sigma^2)(T_2 - T_0)}{\sigma \sqrt{T_2 - T_0}}$$

$$d_2 = d_1 - \sigma \sqrt{T_2 - T_0}$$

The digital option payoff is given by:

$$C_T = I\{S_T > K\}, \quad C_0 = e^{-r_{1,3}(T_3-T_1)} N(d_2) \quad (10)$$

$$P_T = I\{S_T < K\}, \quad P_0 = e^{-r_{1,3}(T_3-T_1)} N(-d_2) \quad (11)$$

Then, estimated value of the replica is

$$V_T = FV_B \cdot e^{-r_{0,3}(T_3-T_1)} + n_1 \cdot C(K_1) - n_2 \cdot C(K_2) + n_3 \cdot P(K_1) - 100 \cdot P^{digital}(K_3) - n_5 \cdot P(K_3) \quad (12)$$

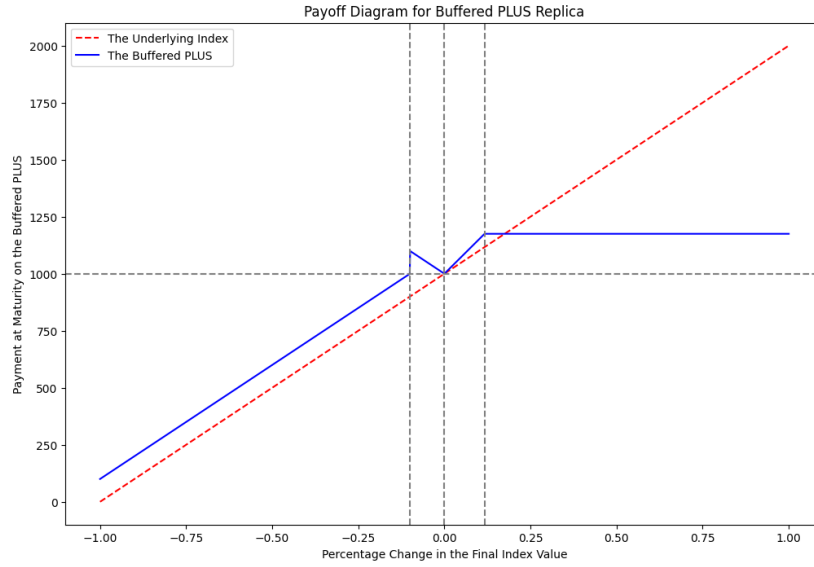


Figure 1: Payoff Diagram for Buffered PLUS Replica

3 Sensitivity Analysis

To assess the impact of key market variables on the valuation of the Buffered PLUS instrument, we conducted a sensitivity analysis by varying the following parameters within reasonable market ranges

3.1 Key market variables value range

- **Historical Volatility (σ):** Ranges from 8% to 20%.

Approximate Vega (price sensitivity to volatility): -223.25

- **Risk-Free Rate (r):** Tested between 1% and 8%.
Approximate sensitivity to riskfree rate: -890.67
- **Dividend Yield (q):** Varies from 0.5% to 3%.
Approximate sensitivity to dividend yield: -929.20
- **Time to Maturity (T):** Ranges from 1 year to 3 years.
Approximate sensitivity to time to maturity: -43.11

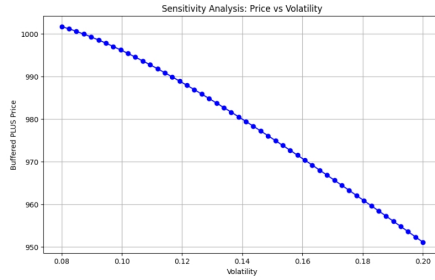


Figure 2: Historical Volatility

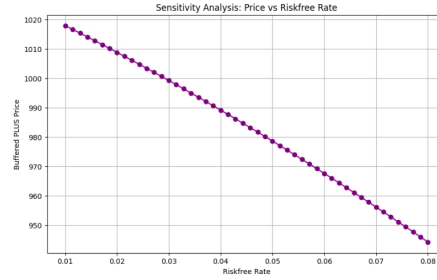


Figure 3: Risk-Free Rate

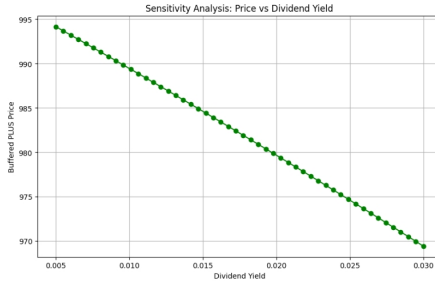


Figure 4: Dividend yield



Figure 5: Time to maturity

3.2 Interpretation of Results

The sensitivity analysis confirms that:

- Higher volatility (σ) increases the option values in the Buffered PLUS, leading to a higher valuation. The Figure 1 shows the product's price declining as implied volatility increases from 8% to 20%. The drop reflects the net short option exposure, which becomes more expensive at higher volatility. The approximate Vega of -223.25 indicates how the price changes for a small volatility shift and thus differs from the overall decrease observed across the full 8%–20% range.

- Higher risk-free rate (r) reduces present values due to discounting effects. This Figure 3 illustrates how the product's price decreases as the risk-free rate rises from 1% to 8%. The drop is primarily driven by the lower present value of the bond component (due to higher discounting) and the net effect on the embedded options. The approximate sensitivity of around -90.67 is a local measure of how the price changes for a very small shift in the rate, which can differ from the overall change of roughly 70 observed over the full 1%–8% range.
- Higher dividend yield (q) lowers call option values, slightly reducing Buffered PLUS valuation. The Figure 4 shows how the product's price declines as the dividend yield increases from 0.5% to 3.0%. The decrease reflects the negative effect of higher dividends on the embedded call options, which become less valuable when dividends rise. The approximate sensitivity of around -929.20 is a local measure for a small change in yield and thus may differ from the total price change observed over the entire range.
- Longer time to maturity (T) affects valuation through time decay and volatility impact on option pricing. The Figure 5 shows how the product's price decreases as time to maturity extends from around 1.0 to 3.0 years, reflecting the combined impact of discounting and net short option exposure. The approximate sensitivity of -43.11 is a local measure for a small shift in maturity, which differs from the overall price change observed across the full range.